

Examining Race in Healthcare-Focused HCI Research

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ABSTRACT

The human-computer interaction community has a long tradition of conducting research in healthcare settings. We conducted a systematic review of how race is defined and discussed in healthcare-focused HCI research. Beginning with an initial set of 514 articles drawn from two major HCI venues, we applied exclusion criteria resulting in a data set of 29 articles. We found considerable variation in definitions of race across articles, with some focusing on physical appearances, others on sociocultural differences, and still others not providing any explicit definition of race. This variation was further reflected in the method used to identify participants' race and the level of specificity used in categorizing participants' race and ethnicity. Additionally, many articles engagement with race and racial disparities in minor ways, with some articles proposing racial equity as an avenue for future work without attending to race in their current investigation. Finally, we found that most articles studied patients and end users, leaving unexamined the practices and perspectives of healthcare practitioners and institutions. We conclude with suggestions advocating greater clarity and consciousness around race, racism, and racial disparities in healthcare focused HCI research.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*; • **Applied computing** → *Health care information systems; Health informatics*.

KEYWORDS

HCI, Healthcare, Race, Critical Analysis, Systematic Review

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1 INTRODUCTION

This paper reports preliminary results from a systematic review of race in healthcare-focused human-computer interaction (HCI) literature. With the Black Lives Matter movement coming to the forefront of media attention following the murder of George Floyd

at the hands of police in the summer of 2020, there has been ongoing public discussion about systemic racism and its widespread effects. Healthcare disparities have been a central focus of these discussions. Racial health disparities are multifaceted; they are the result of structural and systemic factors including wealth gaps, unequal access to education and jobs, and personal bias in the healthcare system; and they have severe consequences for racialized people including systematically higher rates of preventable illness and death across health conditions [3, 7, 24]. The increased attention to racial health disparities in recent public discourse has also been reflected in scholarship in HCI and CSCW, with studies seeking to uncover and address sources of bias and racism in healthcare (e.g., [12, 16, 22, 28, 29]) and raising awareness about racism and critical race theory in HCI and academic research [20].

Alongside the increased attention to racial health disparities, researchers have shown that there is a lack of consensus around the definition of race in medical research [27]. The term race can refer to a variety of things including physical appearance, social category, ethnic background, ancestry, nationality, and country of origin. A lack of precision and clarity around the multiple meanings of race can have serious consequences such as the inability to compare and synthesize findings across studies, and in HCI, mismatches between designer intentions and user needs and desires.

Hence, we sought to understand how race is interpreted and discussed in healthcare-focused scholarship in CSCW and HCI by conducting a systematic literature review. We searched the databases of two major HCI venues for articles that mentioned race in the context of healthcare, resulting in 29 articles after removing articles that did not meet our exclusion criteria. We conducted a thematic analysis of these articles. We found that definitions of race are unclear and inconsistent in HCI healthcare research. Further, we found that many articles fail to substantially engage with race, racism, and racial disparities, with a few articles referring to race only in participant demographics and future work. We conclude with suggestions to foster careful and critical engagement around racial disparities in HCI healthcare research.

2 RELATED WORK

In shining a light on race as a vital dimension of analysis in HCI, we stand on the shoulders of many giants. A detailed review of prior work is outside the scope of this short submission, but we provide a brief overview to establish the background. Harrington et al. conducted a systematic review of HCI research on aging [12]. Based on an examination of 174 articles, they found that HCI research on aging treats white people as the default or baseline and effectively “others” Black people and people of color [12]. Further, they note that race is under-examined in HCI research on aging and older adults, with only four studies examining how “the race of participants as it intersects with technology requirements” [12].

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Systematic reviews of the broader HCI literature have shown that race is consistently and vastly under-reported in comparison to other identity dimensions such as gender and age [12, 14, 26]. We seek to address this gap pointed out by suggesting ways to improve the reporting and analysis of race in HCI healthcare literature.

Healthcare-focused research in CSCW/HCI often seeks to uncover and address sources of bias and racism in healthcare, as the HCI subfield most invested in improving the lives of stakeholders in the healthcare ecosystem. For instance, Veinot et al. demonstrate that healthcare informatics interventions can amplify inequality because they are “more accessible to, heavily used by, adhered to, or effective for those from socioeconomically advantaged groups” [29]. Walker et al. show that incentive structures in the field of public health intensify inequities by discouraging experts from collaborating with Black women and reproductive justice advocates who are not well represented in the public health domain [32]. Pendse et al. show that the standards and artifacts used in digital mental health tools may exacerbate the inequities faced by historically minoritized groups [22]. Despite this excellent and timely work, little attention has been paid to how the term race is defined and used in healthcare-focused subfield of HCI. To our knowledge, we are the first to critically examine race in HCI healthcare research.

3 METHODS

We conducted a systematic review [17] of two major HCI venues, the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW) and the ACM CHI Conference on Human Factors in Computing Systems (CHI). We searched the ACM Digital Library for articles that mentioned race and healthcare. We used queries such as “race AND (medicine OR healthcare)” to ensure that articles dealt with both topics, and filtered articles by proceedings to focus on CHI and CSCW. We included all articles from the results in the first phase of the review (n=514).

After removing duplicate records (n=51), we screened abstracts to narrow the list to articles that explicitly discussed race in the context of healthcare. In the case of articles that did not discuss race or healthcare, we searched for the terms “race,” “racial,” “racism,” “healthcare,” “medicine,” and “medical” in the full text of the article to find where the topic was discussed. Articles were included in the next step if they contained any mention of race and were discussing topics relevant to healthcare. Papers were excluded if they met any of the following exclusion criteria: (a) the paper used the word race in reference to something other than racial categories and racialized interactions, such as car racing games [13]; b) the paper had no connection to the healthcare fields, such as papers dealing solely with law enforcement [18]; c) the paper was not a research study, literature review, or essay, such as the introduction to conference proceedings or workshops [34]; and d) the paper mentioned race and healthcare entirely independently of each other. For instance, a paper on the legalization of same-sex marriage refers separately to healthcare access and racism [2].

This resulted in a set of 29 papers. We conducted a thematic analysis on these articles [6]. We developed a set of themes that were used to code each article including study characteristics (e.g., methods), how race is defined (e.g., social construct versus biological category), how race is operationalized (e.g., the list of categories),

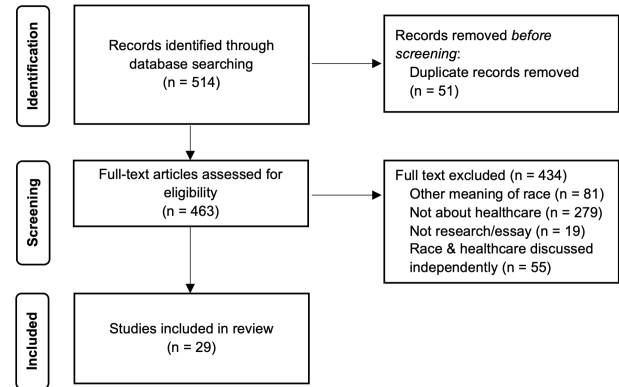


Figure 1: Flowchart of literature selection process according to PRISMA guidelines [21].

and how race appears in the article (e.g., the article meaningfully engages with race throughout versus mentioning it in future work). The first author conducted the coding after reading through each article. Both authors met weekly to discuss newly coded data, clarify thematic categories, and develop emergent observations. Once coding was complete, we analyzed each theme to identify trends across articles. In terms of positionality, we are two researchers with five years of experience working with racialized groups in healthcare settings in the United States. We recognize that our findings will be shaped by our identities and circumstances. The first author is a white cisgender woman who grew up in a suburban working class family in the Midwest. The second author is a disabled transgender non-binary person of color. They grew up in an urban, middle class, and privileged caste family in India. Results from our analysis are laid out in the following sections.

4 FINDINGS

4.1 Study Characteristics

The articles that we examined included empirical studies, literature reviews, and essays. Empirical studies employed a range of methods including interviews, focus groups, photo elicitation, and surveys, with sample sizes ranging from 20-40 participants for interview studies and 42-267 for survey studies. Most studies (n=22) focused on the perspectives of patients and health technology users. Few studies (n=4) examined the perspectives of healthcare practitioners. Some studies sought to uncover and address the needs and challenges of specific minority groups such as immigrant women (n=9), whereas others designed tools to address previously identified needs such as peer support for patients (n=12).

4.2 Unclear and Inconsistent Race Definitions

All articles in our data set directly referenced race or racism, but 72% of articles (n=21) did not define what they meant by race and left the notion of race implicit. Different articles interpreted the term race differently to mean physical appearance, sociocultural background, ethnic identity, nationality, or country of origin without presenting definitions or justifications, suggesting a lack of clarity and consensus around the definition of race in HCI healthcare research.

Most articles ($n=22$) named the racial categories that participants belonged to, but the set of categories used as well as their level of specificity varied between articles. Some articles used categories drawn from the US Census, operationalizing race using the following list: White, Black or African American, American Indian or Alaska Native, Asian, and Native Hawaiian or Other Pacific Islander. In contrast, other articles provided “other” as the only alternative to the White/Caucasian, Black/African American, and Asian categories. Two articles clubbed racial and ethnic categories together, for example using Latinx and Afro-Caribbean as racial categories.

Further, 24% of articles ($n=7$) failed to mention whether participants’ racial identities were self-reported, researcher-assigned, or otherwise inferred. Additionally, two studies used the words race and culture interchangeably, always when referencing minoritized racial identities and non-white or non-western societies (e.g., Afro-Caribbean people). That is, some studies implicitly equate racial difference with cultural difference without justification.

4.3 Limited Engagement with Race and Racism

Many articles engaged with race in a very limited manner. For example, articles mentioned race in passing without further discussion about race, racism, or racial processes in empirical data, data analysis, conceptual theorization, or design (i.e., race was absent in most of the paper). 38% of studies ($n=11$) described racial identity in the context of participant demographics and recruitment, but neither analyzed race as a dimension of difference in the findings nor presented negative findings about the lack of difference along racial lines. This absence was echoed in the discussion, design implications, and subsequent sections of such papers. Studies thus reported the racial makeup of recruited participants without substantially engaging with race, racism, and racial disparities in healthcare.

Further, 14% of articles ($n=4$) proposed racial diversity and greater awareness about racism as goals for future research, without attending to racial dimensions in their own work (i.e., papers did not themselves engage with race but suggested that future researchers should do so). These articles either had an insufficiently diverse sample of participants or developed a general purpose tool (e.g., a virtual nurse) that was hypothesized to be useful in addressing racism. However, these articles left for future researchers the tasks of evaluating the anti-racist potential of such tools and recruiting diverse participants. In a refreshing contrast, ten articles substantively engaged with race and racial disparities.

Finally, 20% of articles ($n=6$) used cultural differences to explain the relationship between race and health disparities. For example, studies discussed how cultural variation in diet, family routines, and views on health management impacted the outcomes of non-white marginalized communities, typically leading to higher rates of illness and mortality. Studies also attributed negative experiences of immigrants of color with the US healthcare system to cultural differences with their country of origin, citing reasons such as differing norms around doctor-patient communication. Thus, these studies held that cultural differences contributed to health disparities.

5 DISCUSSION

We sought to examine how race is used in healthcare-focused HCI research. Based on a thematic analysis of 29 articles, we found

that race is frequently left undefined and there is considerable variation in how race is operationalized across studies. Further, articles failed to substantially engage with race. We briefly discuss the implications of these findings.

5.1 Clarity and Consistency in the Use of Race

We found that the majority of articles did not define what they meant by race and left the notion of race implicit. The absence of definitions for commonly used terms could be part of a larger trend in HCI literature: Stowell and colleagues’ review article on mobile health technology, for example, shows that the literature frequently uses terms like “barriers to [healthcare] access” yet rarely defines the term or specifies the barriers being referred to in a given instance [28]. Missing or implied definitions of key terms require readers to make assumptions that may not always be correct, risking potential misinterpretation of findings and inconsistency in findings across studies. This insight on the lack of a clear definition for race in HCI healthcare literature is consistent with critical analyses of medical literature: sociologists have shown that medical research lacks consensus around the terms race and ethnicity [27]. We recommend that researchers clearly lay out what they mean by race and other important terms, especially when definitions can vary as with race (which can refer to appearance, social grouping, ethnic background, ancestry, nationality, and country of origin). We join recent calls from medical literature suggesting that “defining race is essential to interpret individual studies and to facilitate comparisons” [33]. That is, clear definitions of race would allow researchers to compare and synthesize across the many HCI healthcare studies that are beginning to examine racism and disparities.

Participants’ racial identities can be self reported, researcher identified, or inferred from other information such as an established database. We found that several articles failed to specify the mode of data collection used to obtain participant demographics. Best practice guidelines suggest that race information should be gathered via self-report to ensure that it accurately reflects participant identity [33]. Given the variability in definitions of race across articles, we suggest that studies include information about how demographic data was collected to clarify the specific operationalization of race used in a given study and appropriately contextualize its findings. When analyzing findings or comparing results across studies, we propose that researchers recognize that race is an evolving and fluid construct that may be defined differently in differing contexts and times [4, 23]. Finally, we urge researchers to consider whether racial identity information is relevant to the questions they seek to answer. Collecting racial identity information without justification can have negative consequences: cross-group comparisons can stigmatize individual communities (instead of focusing on systemic inequities), participants from smaller populations risk exposure and re-identification, and identity information can be misused for profiling and discrimination [8–11, 15, 19].

5.2 Engage with Race and Health Disparities

In our data set, it was common to see minor mentions of race in participant demographics and as an avenue for future work. Given the widespread racial disparities in the healthcare system, we suggest that the analysis should go further and examine racism, racial

disparities, and race as a category of human difference in greater depth. Without dedicating substantial effort and space in the article towards such analyses, the inclusion of racial identity data in demographics and minor mentions in future work seem performative and disingenuous, because they make empty gestures towards racial diversity and brush off the topic of race for later investigation. We suggest that there needs to be higher race consciousness in healthcare-focused HCI research. Perhaps a first step would be to educate ourselves about structural and interpersonal racism in healthcare and society both historically and in the present [3, 5, 24, 30, 31].

Socioeconomically disadvantaged racial groups are adversely impacted by health disparities, i.e., higher rates of illness and death, than the rest of the population. We found that several articles in our data set assumed that cultural differences led to health disparities. This conceptualization is a grave misconception that ignores the effects of systemic racism that systematically produces obstacles such as “social, economic, and/or environmental disadvantage” faced by socioeconomically disadvantaged racial groups [7, 30]. Further, articles that emphasized cultural differences tended to use white people and mainstream American culture as the reference group [12], implying that the cultures of non-white people are inherently different from white people, outside the mainstream, and un-American. This deeply problematic other-ization of non-white people ignores the fact that “racism – not race – drives health disparities” [1, 30].

5.3 Examine Expert Practice and Institutions

We found that most articles focused on the perspectives of patients and health technology users, indicating a focus on individual challenges rather than systemic challenges or challenges faced at the level of practitioners and institutions. Race-conscious HCI literature naturally centers patients and communities as groups that are most vulnerable to harm by flawed systems, a focus likely arising from the user-centered paradigm in HCI and the patient-centered approach in healthcare. However, it also points to the need for race-conscious research that generates implications for practitioners and institutions. For example, HCI has a long tradition of studying the work practices of healthcare practitioners, but our review only found four such articles. This indicates a need for more attentiveness to issues of race when studying practitioners and institutions. Studies from adjacent fields provide examples of efforts to investigate systemic issues, such as a health services research study examining how practitioner and institutional biases contribute to disparities in diagnostic imaging based on an analysis of over 300,000 emergency room visits [25]. While community based approaches help understand minoritized groups’ experiences, we propose that HCI healthcare scholars must also examine expert practice and institutional processes to advance the goal of addressing systemic racism.

5.4 Limitations and Future Work

Our observations are limited to two major venues in the field of HCI (CHI and CSCW). Given the high selectivity (i.e., low acceptance rates) of these venues, our hope was to examine how race is represented in highly regarded HCI healthcare research. However, we did not analyze articles from other venues and so our findings may not extend to other HCI venues and fields. Further, we prioritized

articles mentioning race in the context of medicine and healthcare, and as a result our findings may not generalize to HCI research outside healthcare settings. We hope to broaden our analysis in the future, and suggest that future work should take a broader view of the literature by incorporating other venues and repositories (such as the ACM Digital Library and IEEE Xplore Digital Library). Future research should also draw inspiration from domains outside healthcare that engage substantively with race and racism.

6 CONCLUSION

We systematically reviewed healthcare-focused literature in proceedings of the ACM CHI and CSCW conferences. We found that race was left undefined in many articles and inconsistently operationalized across articles. Further, we found that articles did not always engage meaningfully with race, despite including equity considerations in recruitment and future work. We conclude with suggestions to help future HCI healthcare research engage substantively with race.

REFERENCES

- [1] Bridget Balch. 2022. Racism – not race – drives health disparities. *Association of American Medical Colleges News* (Nov. 2022). <https://www.aamc.org/news/racism-not-race-drives-health-disparities>
- [2] Jeffrey Bardzell, Guo Freeman, Shaowen Bardzell, and Pei-Ying Chen. 2020. Join.Love: A Sociotechnical Genealogy of the Legalization of Same-Sex Marriage. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376603>
- [3] Ruha Benjamin. 2019. *Race After Technology: Abolitionist Tools for the New Jim Code*.
- [4] Geoffrey C Bowker and S. Leigh Star. 1999. *Sorting Things Out: Classification and its consequences*. Cambridge, MA. <http://mitpress.mit.edu/books/sorting-things-out> Publication Title: MIT Press.
- [5] Lundy Braun. 2014. *Breathing race into the machine : the surprising career of the spirometer from plantation to genetics*. University of Minnesota Press.
- [6] V. Braun and V. Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3 (2006), 77–101. <https://doi.org/10.1191/1478088706qp0630a> arXiv: 1011.1669v3 ISBN: 1478-0887\1478-0895.
- [7] Centers for Disease Control and Prevention. 2017. Health Disparities. <https://www.cdc.gov/aging/disparities/index.htm>
- [8] Gene Demby. 2014. On The Census, Who Checks 'Hispanic,' Who Checks 'White,' And Why. *NPR* (June 2014). <https://www.npr.org/sections/codeswitch/2014/06/16/321819185/on-the-census-who-checks-hispanic-who-checks-white-and-why>
- [9] Brian Gratton and Emily Klancher Merchant. 2016. La Raza: Mexicans in the United States Census. *Journal of Policy History* 28, 4 (2016), 537–567. <https://doi.org/10.1017/S0898030616000257> Publisher: Cambridge University Press.
- [10] Sujata Gupta. 2020. To fight discrimination, the U.S. census needs a different race question. *ScienceNews* (March 2020). <https://www.sciencenews.org/article/census-2020-race-ethnicity-questions>
- [11] Rochelle Gutiérrez. 2008. A “Gap-Gazing” Fetish in Mathematics Education? Problematising Research on the Achievement Gap. *Journal for Research in Mathematics Education* 39, 4 (2008), 357–364. <http://www.jstor.org/stable/40539302>
- [12] Christina Harrington, Aqueasha Martin-Hammond, and Kirsten E Bray. 2022. Examining Identity as a Variable of Health Technology Research for Older Adults: A Systematic Review. In *CHI Conference on Human Factors in Computing Systems*. ACM, New Orleans LA USA, 1–24. <https://doi.org/10.1145/3491102.3517621>
- [13] Richard L. Hazlett. 2006. Measuring emotional valence during interactive experiences: boys at video game play. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (CHI '06). Association for Computing Machinery, New York, NY, USA, 1023–1026. <https://doi.org/10.1145/1124772.1124925>
- [14] Julia Himmelsbach, Stephanie Schwarz, Cornelia Gerdenitsch, Beatrix Wais-Zechmann, Jan Bobeth, and Manfred Tscheligi. 2019. Do We Care About Diversity in Human Computer Interaction: A Comprehensive Content Analysis on Diversity Dimensions in Research. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3290605.3300720>
- [15] Farah Kader, Lan N Doan, Matthew Lee, Matthew K Chin, Simona C Kwon, and Stella S Yi. 2022. Disaggregating Race/Ethnicity Data Categories: Criticisms,

- Dangers, And Opposing Viewpoints. 10.1377/forefront.20220323.555023
- [16] Elizabeth Kaziunas, Michael S. Klinkman, and Mark S. Ackerman. 2019. Precarious Interventions: Designing for Ecologies of Care. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW (Nov. 2019). <https://doi.org/10.1145/3359215> Place: New York, NY, USA Publisher: Association for Computing Machinery.
 - [17] Barbara Kitchenham, O. Pearl Brereton, David Budgen, Mark Turner, John Bailey, and Stephen Linkman. 2009. Systematic literature reviews in software engineering – A systematic literature review. *Information and Software Technology* 51, 1 (2009), 7–15. <https://doi.org/10.1016/j.infsof.2008.09.009> Special Section - Most Cited Articles in 2002 and Regular Research Papers.
 - [18] Alex Mariakakis, Sayna Parsi, Shwetak N. Patel, and Jacob O. Wobbrock. 2018. Drunk User Interfaces: Determining Blood Alcohol Level through Everyday Smartphone Tasks. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, Montreal QC Canada, 1–13. <https://doi.org/10.1145/3173574.3173808>
 - [19] JR Minkel. 2007. Confirmed: The U.S. Census Bureau Gave Up Names of Japanese-Americans in WW II. *Scientific American* (March 2007). <https://www.scientificamerican.com/article/confirmed-the-us-census-b/>
 - [20] Ihudiya Finda Ogbonnaya-Ogburu, Angela D.R. Smith, Alexandra To, and Kentaro Toyama. 2020. Critical Race Theory for HCI. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '20*). Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3313831.3376392>
 - [21] Matthew J Page, Joanne E McKenzie, Patrick M Bossuyt, Isabelle Boutron, Tammy C Hoffmann, Cynthia D Mulrow, Larissa Shamseer, Jennifer M Tetzlaff, Elie A Akl, Sue E Brennan, Roger Chou, Julie Glanville, Jeremy M Grimshaw, Asbjørn Hróbjartsson, Manoj M Lalu, Tianjing Li, Elizabeth W Loder, Evan Mayo-Wilson, Steve McDonald, Luke A McGuinness, Lesley A Stewart, James Thomas, Andrea C Tricco, Vivian A Welch, Penny Whiting, and David Moher. 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 372 (2021). <https://doi.org/10.1136/bmj.n71> arXiv:<https://www.bmj.com/content/372/bmj.n71.full.pdf>
 - [22] Sachin R Pendse, Daniel Nkemelu, Nicola J Bidwell, Sushrut Jadhav, Soumitra Pathare, Munmun De Choudhury, and Neha Kumar. 2022. From Treatment to Healing: Envisioning a Decolonial Digital Mental Health. In *CHI Conference on Human Factors in Computing Systems*. ACM, New Orleans LA USA, 1–23. <https://doi.org/10.1145/3491102.3501982>
 - [23] Paola Ramos. 2020. *Finding Latinx: In Search of the Voices Redefining Latino Identity*. Penguin Random House.
 - [24] Dorothy Roberts. 2012. *Fatal Invention : How Science, Politics, and Big Business Re-create Race in the Twenty-First Century*. The New Press, New York, NY, USA.
 - [25] Andrew B. Ross, Vivek Kalia, Brian Y. Chan, and Geng Li. 2020. The influence of patient race on the use of diagnostic imaging in United States emergency departments: data from the National Hospital Ambulatory Medical Care survey. *BMC Health Services Research* 20, 1 (Sept. 2020), 840. <https://doi.org/10.1186/s12913-020-05698-1>
 - [26] Ari Schlesinger, W. Keith Edwards, and Rebecca E. Grinter. 2017. Intersectional HCI: Engaging Identity through Gender, Race, and Class. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (*CHI '17*). Association for Computing Machinery, New York, NY, USA, 5412–5427. <https://doi.org/10.1145/3025453.3025766>
 - [27] Sachil Singh and Valerie Steeves. 2020. The contested meanings of race and ethnicity in medical research: A case study of the DynaMed Point of Care tool. *Social Science & Medicine* 265 (2020), 113112. <https://doi.org/10.1016/j.socscimed.2020.113112>
 - [28] Elizabeth Stowell, Mercedes C. Lyson, Herman Saksono, René C. Wurth, Holly Jimison, Misha Pavel, and Andrea G. Parker. 2018. Designing and Evaluating mHealth Interventions for Vulnerable Populations: A Systematic Review. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (*CHI '18*). Association for Computing Machinery, New York, NY, USA, 1–17. <https://doi.org/10.1145/3173574.3173589>
 - [29] Tiffany C. Veinot, Hannah Mitchell, and Jessica S. Ancker. 2018. Good intentions are not enough: how informatics interventions can worsen inequality. *Journal of the American Medical Informatics Association : JAMIA* 25, 8 (Aug. 2018), 1080–1088. <https://doi.org/10.1093/jamia/ocy052> Place: England.
 - [30] Linda Villarosa. 2022. *Under the skin : the hidden toll of racism on American lives and on the health of our nation* (first edition ed.). Doubleday. 269 pages.
 - [31] Darshali A Vyas, Leo G Eisenstein, and David S Jones. 2020. Hidden in Plain Sight — Reconsidering the Use of Race Correction in Clinical Algorithms. *New England Journal of Medicine* 383, 9 (June 2020), 874–882. <https://doi.org/10.1056/NEJMms2004740> Publisher: Massachusetts Medical Society.
 - [32] Ashley Marie Walker, Michael A. DeVito, Kathryn E. Ringland, and Madhu Reddy. 2019. (In)Visible Choices: Articulation Work and the Rise in US Maternal Mortality. In *Conference Companion Publication of the 2019 on Computer Supported Cooperative Work and Social Computing* (Austin, TX, USA) (*CSCW '19*). Association for Computing Machinery, New York, NY, USA, 403–407. <https://doi.org/10.1145/3311957.3359463>
 - [33] Margaret A. Winker. 2004. Measuring Race and Ethnicity: Why and How? *JAMA* 292, 13 (2004), 1612–1614. <https://doi.org/10.1001/jama.292.13.1612>
 - [34] Marisol Wong-Villares, Aakash Gautam, Wendy Roldan, Lucy Pei, Jessa Dickinson, Azra Ismail, Betsy DiSalvo, Neha Kumar, Tammy Clegg, Sheena Erete, Emily Roden, Nithya Sambasivan, and Jason Yip. 2020. From Needs to Strengths: Operationalizing an Assets-Based Design of Technology. In *Conference Companion Publication of the 2020 on Computer Supported Cooperative Work and Social Computing*. ACM, Virtual Event USA, 527–535. <https://doi.org/10.1145/3406865.3418594>